

MATH 213 – Homework 8

- **Read •Chapter 4 of our textbook, following the guidance below. Answer reading questions (a) – (d). You can keep your answers short and to-the-point. One sentence is all that will be needed for most of these.**
- **Read Section 4.1 on contingency tables and bar plots.**
- **Read Section 4.2.1 on bar plots with two variables.**
- **The authors distinguish between stacked bar plots, dodged bar plots, and standardized bar plots (proportions instead of counts).**
- **The textbook authors offer some advice on when it is appropriate to use a stacked bar plot vs a dodged bar plot. Summarize their advice below, in a few sentences.**

Stacked bar plots show overall distributions and common combinations; dodged bar plots are better for comparing proportions across levels.

- **Section 4.2.2 on Mosaic Plots is optional reading.**
- **Read Section 4.3 on Row and Column Proportions.**
- **From Table 4.3, describe in words what the 0.635 represents. How was this value computed?**

0.635 represents the proportion of joint applicants who have a mortgage. It is computed by dividing the count of joint applicants with a mortgage by the total number of joint applicants.

$$950/1495 = 0.635$$

- **From Table 4.4, describe in words what the 0.135 represents. How was the value computed?**

0.135 represents the proportion of homeowners(own) who applied as joint applicants. This is computed by dividing the count of joint applicants who own their home by the total number of homeowners.

$$183/13530 = .135 \text{ or } 13.5\% \text{ of homeowners applied jointly for a loan.}$$

- **Skim 4.4 about Pie Charts.**

- **What is the main takeaway the authors want you to have about pie charts?**

The primary take away from the section is that it is better to use a pie chart if the goal is to provide a high-level overview and visualize a categorical variable with few levels.

- Question 2 from the • Chapter 4 exercises.
- Write your answers in complete sentences, justifying your responses appropriately.
- Be prepared to discuss your answer to part (e), especially, next time.
- **Notice that part (f) is asking you to suggest a confounding variable. Make sure the variable you suggest IS A VARIABLE for the given cases. ALSO make sure to explain how your potential confounding variable is connected to the given variables!**

a)  $372/910 * 100 = 40.9\%$

40.9% of voters identify as conservatives.

b)  $278/910 * 100 = 30.5$

About 30.5% of the voters are in favor of the citizenship option.

c)  $57/910 * 100 = 6.3\%$

About 6.3% of the voters are conservatives and support the citizenship option

d)

- Conservatives:  $57/372 * 100 = 15.3\%$

About 15.3 % of conservatives support the citizenship option.

- Moderates:  $120/363 * 100 = 33.1\%$

About 33.1% of moderates support the citizenship option.

- Liberals:  $101/175 * 100 = 57.7\%$

About 57.7% of liberals support the citizenship option.

e) Yes, political ideology and views on immigration appear to be associated. The percentages of support for citizenship option vary significantly across political views. For example, liberals (57.7%) are much more likely to support the citizenship option compared to moderates (33.1%) and conservatives (15.3%) this variation suggests that political ideology can influence views on immigration.

f)

One potential confounding variable is the level of education. Higher education might correlate with more liberal political views and more support for immigration policies. Whereas standard education might correlate with conservative views and less support for such policies. Education often shapes individuals' perspective on social issues, a big one is immigration as highlighted in this example this might explain the association above.

- **In Homework 7, you analyzed the association between a person's being married and higher education levels using data from the American Community Survey. We discussed in class some possible confounding variables to consider when analyzing the association between these variables. Since the ACS is an observational study, we would not claim immediately that being married is causing people to get more education.**

**Here is the graph that we studied in Homework 7 to see the association between being married and higher levels of education.**

**One of the possible confounding variables that we identified was age. If we create a new "age group" categorical variable, we can make a big graph comparing education levels for the married and unmarried people in each age group.**

**\* The graph did not transfer over when I converted this to a word document\***

**When I look at the two graphs, I conclude the following:**

**Looking at the whole ACS sample, I see a strong association between being married and higher levels of education. When I compare people in the same age groups, I still see this association, but it is weaker.**

**Write a paragraph to support my claims, using numbers drawn from the two graphs. Make certain to address and to support all three of my points! (Strong association without age groups, association with age groups, but weaker.)**

The graph above, from the 2012 ACS survey, compares the distribution of education levels (graduate, college, and high school or lower) among individuals who are married or not, before factoring in age we can see a strong correlation between individuals and their education level between the two groups. However, after considering age as a variable. While the association between marital status and education level remains visible, it is notably weaker. For instance, in the age range of 26 to 40, about 20 percent of married individuals have graduate degrees, and 35 percent have a college education. In contrast, within the same age range for unmarried individuals, about 10 percent hold graduate degrees, and just over 25 percent have college degrees.

This variation in the strength of association across different age groups highlights the role of age as a confounding factor. It suggests that while being married is associated with higher educational levels, the strength of this association is moderated when the age of the individuals is considered.